1. What is the Fibonacci series?

A series in which each number is the sum of the previous two numbers, starting from 0 and 1.

Example: 0, 1, 1, 2, 3, 5, 8, 13...

2. What is the aim of this program?

To implement Fibonacci numbers using recursive and non-recursive methods and compare their time and space complexities.

3. What is recursion?

Recursion is a technique in which a function calls itself to solve a problem.

4. What is iteration?

Iteration means repeating statements using loops such as for and while.

5. What is the difference between recursion and iteration?

- Recursion uses function calls.
- Iteration uses loops.
- Recursion requires stack memory.
- Iteration generally uses less memory.

6. What is the base condition in your recursive function?

if n <= 1: return n

It stops the recursive function from calling itself further.

7. What is the recurrence relation of Fibonacci?

$$F(n)=F(n-1)+F(n-2)$$

Where F(0) = 0 and F(1) = 1.

8. What is the time complexity of recursive Fibonacci?

$O(2^n)$, because it makes repeated recursive calls.

9. What is the space complexity of recursive Fibonacci?

O(n), because of the recursion call stack.

10. What is the time complexity of iterative Fibonacci?

O(n), because the loop executes approximately n times.

11. What is the space complexity of iterative Fibonacci using two variables?

O(1), because it uses only two variables to calculate the next number.

12. Which method is faster?

The iterative method is faster because it avoids repeated calculations.

13. Why is recursion slower?

Because it calculates the same Fibonacci values multiple times.

14. What is call_count used for?

It counts how many times the recursive function is called.

15. What is time.perf_counter()?

It measures the execution time of the program with high precision.

16. What is the use of global call_count?

It allows the function to modify the global variable call_count.

17. What is the use of range(2, n+1)?

It generates numbers from 2 to n, including n.

18. What is the use of assert v1 == v2?

It checks whether both iterative and recursive methods produce the same result.

19. What is dynamic programming?

Dynamic programming is a technique that stores previously calculated results to avoid repeated calculations.

20. Which method uses an array?

fib_series_iterative() uses an array (Python list) to store Fibonacci numbers.

21. What happens if n = 0?

The Fibonacci value is 0.

22. What happens if n = 1?

The Fibonacci value is 1.

23. Why do we compare time complexity?

To understand which algorithm performs better as the input size increases.

24. What is the advantage of the iterative method?

It is faster and uses less memory when implemented with two variables.

25. What is the conclusion of this program?

Both methods generate the same Fibonacci series, but the iterative method is more efficient in terms of execution time and memory.

Important complexities to remember

Method	**Time**	**Space**
Recursive	$O(2^n)$	O(n)

Iterative with array	O(n)	O(n)
Iterative with 2 variables	O(n)	O(1)

2

1. What is Huffman Encoding?

It is a lossless data compression technique that assigns variable-length binary codes based on character frequency.

2. What is a greedy algorithm?

An algorithm that makes the best possible choice at each step.

3. Why is Huffman Encoding called greedy?

Because it repeatedly selects and merges the two nodes with the smallest frequencies.

4. What is the use of Huffman Encoding?

To compress data and reduce storage space.

5. What is a Huffman tree?

A binary tree used to generate binary codes for characters.

6. What is a min-heap?

A data structure in which the smallest element is always at the root.

7. Why do we use heapq?

To efficiently find the two smallest frequencies.

8. What is the use of Counter?

To count the frequency of each character in a string.

9. What is a prefix code?

A code in which no character's code is the prefix of another character's code.

10. What is the use of generate_codes()?

It generates binary codes by traversing the Huffman tree.

11. What is the use of encode()?

It converts the original message into binary codes.

12. What is the use of decode()?

It converts the encoded message back into the original text.

13. What is the time complexity of Huffman Encoding?

O(k log k), where k is the number of unique characters.

14. What is lossless compression?

Compression in which the original data can be recovered without losing information.

15. What is the space complexity?

O(k), where k is the number of unique characters.

3

1. What is the 0/1 Knapsack problem?

It is an optimization problem where we select items to maximize total value without exceeding the given capacity.

2. Why is it called 0/1 Knapsack?

Because each item can either be selected (1) or rejected (0). We cannot take a fraction of an item.

3. What is Dynamic Programming?

It is a technique that solves a problem by breaking it into smaller subproblems and storing their results.

4. What is the aim of this program?

To find the maximum value of selected items within the given knapsack capacity.

5. What is the recurrence relation?

\[T[i][j]=\max(T[i-1][j],V_i+T[i-1][j-W_i]) \]

When the item fits in the remaining capacity.

6. What is the time complexity?

O(n × W), where n is the number of items and W is the capacity.

7. What is the space complexity?

O(n × W), because we store a 2D DP table.

8. What is the use of the DP table?

It stores the maximum value possible for each number of items and capacity.

9. What happens if an item exceeds the capacity?

That item is not selected, and the previous maximum value is retained.

10. What is the difference between 0/1 Knapsack and Fractional Knapsack?

- 0/1 Knapsack: Items cannot be divided.

- Fractional Knapsack: Items can be divided into fractions.

11. Why do we initialize the DP table with zeros?

Because when there are no items or the capacity is zero, the maximum value is zero.

12. What is backtracking in this program?

It is used to trace the DP table backward to identify which items were selected.

13. What is the use of max()?

It selects the maximum value between including and excluding an item.

14. What is the difference between Greedy and Dynamic Programming?

Greedy makes the best local choice at each step, while Dynamic Programming considers and stores solutions to subproblems.

15. What is the conclusion of this program?

The 0/1 Knapsack problem is successfully solved using Dynamic Programming to obtain the maximum value within the given capacity

4

1. What is the N-Queens problem?

The N-Queens problem is placing N queens on an N × N chessboard so that no two queens attack each other.

2. What is backtracking?

Backtracking is an algorithmic technique that tries possible solutions and goes back when a solution becomes invalid.

3. Why do we use backtracking in N-Queens?

Because it checks safe positions and removes a queen when no valid position is available.

4. What are the attacking conditions of a queen?

A queen can attack horizontally, vertically and diagonally.

5. What is the aim of this program?

To place N queens on a chessboard using backtracking, with the first queen already placed at a given position.

6. What is the use of is_safe()?

It checks whether placing a queen at a particular position is safe.

7. What is the use of solve()?

It places queens row by row using recursion and backtracking.

8. What is the use of board?

It stores the column position of each queen in its corresponding row.

9. What does -1 represent in the board?

It represents a row where no queen has been placed.

10. What is the base condition in this program?

if row == n: return True

It means all queens have been successfully placed.

11. What is the use of board[row] = -1?

It removes the queen when a placement does not lead to a valid solution. This is called backtracking.

12. What is the time complexity of N-Queens?

O(N!) in the worst case for this row-by-row backtracking approach.

13. What is the space complexity?

O(N), for the board array and recursion stack.

14. Why is the first queen placed before calling solve()?

Because the program is designed to find a solution with the first queen fixed at a given position.

15. What is the conclusion of this program?

The N-Queens problem is successfully solved using backtracking by placing queens so that no two queens attack each other.

5

1. What is Quick Sort?

Quick Sort is a sorting algorithm that uses the divide-and-conquer technique to sort elements.

2. What is the divide-and-conquer technique?

It divides a problem into smaller subproblems, solves them and combines the results.

3. What is a pivot?

A pivot is an element used to divide the array into two parts.

4. What is deterministic Quick Sort?

Quick Sort in which the pivot is selected using a fixed rule. In this program, it is always the last element.

5. What is randomized Quick Sort?

Quick Sort in which the pivot is selected randomly from the given sub-array.

6. What is the difference between deterministic and randomized Quick Sort?

- Deterministic: Selects the last element as pivot.
- Randomized: Selects a random element as pivot.

7. What is partitioning?

Partitioning rearranges elements so that smaller elements come before the pivot and larger elements come after it.

8. What is the time complexity of Quick Sort?

Case	**Time Complexity**
Best	O(n log n)
Average	O(n log n)
Worst	$O(n^2)$

9. When does the worst case occur?

When the pivot repeatedly becomes the smallest or largest element, creating highly unbalanced partitions.

10. What is the expected time complexity of randomized Quick Sort?

O(n log n).

11. What is the use of random.randint()?

It selects a random pivot index between the lower and upper bounds.

12. What is the use of comparisons?

It counts the number of element comparisons performed during partitioning.

13. What is the use of time.perf_counter()?

It measures the execution time of the sorting algorithm.

14. Is Quick Sort stable?

No. Quick Sort is generally not a stable sorting algorithm.

15. What is the space complexity of Quick Sort?

- Average: O(log n)
- Worst: O(n)

ML

What is the aim of this experiment?
To predict Uber ride fares using Linear Regression and Random Forest Regression.

2. Which type of Machine Learning is used?
Supervised Learning.

3. What is Linear Regression?
An algorithm used to predict continuous numerical values using a linear relationship.

4. What is Random Forest Regression?
An algorithm that combines multiple decision trees to predict numerical values.

5. What is the difference between Linear Regression and Random Forest?
Linear Regression fits a straight line, while Random Forest uses multiple decision trees.

6. What is the target variable?
Uber ride fare (price).

7. What are the input features?
Pickup latitude, pickup longitude, drop-off latitude, drop-off longitude, and passenger count.

8. Why is data preprocessing required?
To handle missing values, remove incorrect data, and prepare data for training.

9. What is train-test split?
Dividing the dataset into training and testing data.

10. What is the use of fit() and predict()?
fit() trains the model, while predict() predicts the output.

11. What is regression?
Regression predicts continuous numerical values.

12. What is MAE?
Mean Absolute Error measures the average absolute difference between actual and predicted values.

13. What is MSE?
Mean Squared Error measures the average squared prediction error.

14. What is R^2 score?
It measures how well the model explains variations in the target variable. Closer to 1 is generally better.

15. Which model is better for Uber fare prediction?
The model with lower MAE and RMSE and a higher R^2 score.

16. What is overfitting?
When a model performs well on training data but poorly on testing data.

17. Why do we use Random Forest?
It can capture complex relationships and reduce overfitting compared with a single decision tree.

18. Which Python libraries are used?
Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

19. What is the difference between classification and regression?
Classification predicts categories, while regression predicts numerical values.

20. What is the output of this experiment?
The predicted Uber ride fare based on pickup and drop-off details.

2

1. What is the aim of this experiment?
To classify emails as Spam or Not Spam using KNN and SVM algorithms.

2. What is binary classification?
Classification that divides data into two classes.

3. What are the two classes in this experiment?

- Normal: Not Spam (0)
- Abnormal: Spam (1)

4. What is KNN?
K-Nearest Neighbors is an algorithm that classifies data based on the majority class of its nearest neighbors.

5. What is SVM?
Support Vector Machine is an algorithm that finds the best decision boundary to separate classes.

6. What is the difference between KNN and SVM?
KNN uses nearby data points for classification, while SVM uses an optimal separating boundary.

7. What is the value of K in KNN?
K represents the number of nearest neighbors considered for classification.

8. What is a Support Vector?
Data points closest to the decision boundary that help define the boundary.

9. What is a decision boundary?
A line or surface that separates different classes.

10. What is feature extraction?
Converting email text into numerical features that a machine learning model can understand.

11. Why is preprocessing required?
To clean email text and convert it into a suitable format for classification.

12. What is TF-IDF?
Term Frequency–Inverse Document Frequency converts text into numerical values based on word importance.

13. What is a confusion matrix?
A table used to evaluate classification performance using TP, TN, FP, and FN.

14. What is accuracy?
The percentage of correctly classified predictions.

15. What is precision?
The proportion of predicted spam emails that are actually spam.

16. What is recall?
The proportion of actual spam emails correctly identified.

17. What is F1-score?
The harmonic mean of precision and recall.

18. Why do we use a train-test split?
To train the model on one part of the dataset and test its performance on unseen data.

19. Which algorithm is better, KNN or SVM?
The algorithm with better evaluation results is preferred. SVM often performs well on high-dimensional text data.

20. What is the final output?
The email is classified as Spam or Not Spam.

3

. What is the aim of this experiment?
To predict whether a bank customer will leave the bank within the next 6 months using a Neural Network.

2. What is Customer Churn?
Customer churn means a customer stops using the bank's services.

3. Which type of Machine Learning is used?
Supervised Learning.

4. What type of problem is this?
Binary Classification.

5. What are the two output classes?

- 0 – Customer will stay.
- 1 – Customer will leave.

6. What is a Neural Network?
A machine learning model inspired by the human brain that learns patterns from data.

7. What are the main layers of a Neural Network?
Input layer, hidden layer, and output layer.

8. What is an activation function?
It introduces non-linearity into a neural network.

9. Which activation functions are commonly used?
ReLU for hidden layers and Sigmoid for binary classification output.

10. What is the purpose of the Sigmoid function?
It converts the output into a probability between 0 and 1.

11. What are the input features?
Customer age, gender, geography, credit score, balance, salary, tenure, and number of products.

12. What is the target variable?
Whether the customer leaves the bank or stays.

13. What is data normalization?
Scaling numerical values to a common range so that features with larger values do not dominate learning.

14. What is an epoch?
One complete pass through the training dataset.

15. What is backpropagation?
The process of calculating errors and updating weights to improve the model.

16. What is a loss function?
It measures the difference between actual and predicted outputs. Binary Cross-Entropy is commonly used for binary classification.

17. What is an optimizer?
It updates model weights to minimize the loss. Adam is a commonly used optimizer.

18. What is overfitting?
When a model performs well on training data but poorly on unseen data.

19. How do you evaluate the model?
Using accuracy, precision, recall, F1-score, and confusion matrix.

20. What is the final output?
The model predicts whether a customer is likely to stay or leave the bank.

4

1. What is the aim of this experiment?
To implement the K-Nearest Neighbors algorithm on the diabetes dataset and evaluate its performance.

2. What is KNN?
K-Nearest Neighbors is a supervised machine learning algorithm used for classification and regression.

3. Which type of learning is KNN?
Supervised Learning.

4. What is the diabetes dataset used for?
To predict whether a person has diabetes or not based on medical features.

5. What is the target variable?
Outcome: 0 means non-diabetic and 1 means diabetic.

6. What is the value of K?
K is the number of nearest neighbors considered for prediction.

7. How does KNN work?
It calculates distances, selects the K nearest neighbors, and predicts based on majority voting.

8. Which distance metric is commonly used in KNN?
Euclidean distance.

9. Why is feature scaling important in KNN?
Because KNN is distance-based, scaling prevents large-valued features from dominating distance calculations.

10. What is a confusion matrix?
A table that compares actual and predicted classes.

11. What is accuracy?
The percentage of correctly classified predictions.

12. What is error rate?
The percentage of incorrectly classified predictions.

13. What is precision?
The proportion of predicted positive cases that are actually positive.

14. What is recall?
The proportion of actual positive cases correctly predicted.

15. What is the use of train_test_split()?
It divides the dataset into training and testing data.

16. What is the use of fit()?
It trains the KNN model using training data.

17. What is the use of predict()?
It predicts the output for test data.

18. What happens if K is too small?
The model may become sensitive to noise and overfit.

19. What happens if K is too large?
The model may underfit and produce less accurate predictions.

20. Which Python library is used to implement KNN?
Scikit-learn (sklearn).

. What is the aim of this experiment?
To group customers based on sales data using K-Means or Hierarchical Clustering.

2. What is clustering?
Clustering is an unsupervised learning technique that groups similar data points together.

3. What is unsupervised learning?
Learning from data without predefined labels.

4. What is K-Means clustering?
An algorithm that divides data into K clusters based on similarity.

5. What is a centroid?
The center point of a cluster.

6. How does K-Means work?

- Select K initial centroids.
- Assign data points to the nearest centroid.
- Recalculate centroids.
- Repeat until clusters become stable.

7. What is the elbow method?
A technique used to determine the optimal number of clusters.

8. What is WCSS?
Within-Cluster Sum of Squares measures the sum of squared distances between data points and their cluster centroids.

9. Why is the elbow method used?
To find the optimal K value by identifying the point where WCSS starts decreasing slowly.

10. What is Hierarchical Clustering?
An unsupervised algorithm that creates a hierarchy of clusters.

11. What are the types of Hierarchical Clustering?
Agglomerative (bottom-up) and Divisive (top-down).

12. What is a dendrogram?
A tree-like diagram that shows how clusters are merged or divided.

13. What is the difference between K-Means and Hierarchical Clustering?
K-Means requires the number of clusters in advance, while Hierarchical Clustering can be visualized using a dendrogram to choose clusters.

14. Why is feature scaling important?
To ensure features with larger values do not dominate distance calculations.

15. Which distance metric is commonly used?
Euclidean distance.

16. What is the use of fit()?
It trains the clustering algorithm on the dataset.

17. What is the use of fit_predict()?
It fits the model and returns cluster labels for each data point.

18. Is clustering supervised or unsupervised?
Unsupervised learning.

19. What is the use of StandardScaler?
It standardizes features by scaling them to mean 0 and standard deviation 1.

20. What is the final output?
Customers are grouped into different clusters based on similarities in sales data.